

Longitudinal Data Analysis with R Statistical Software

Day 2: Generalized Estimating Equations

Benjamin French and Bryan Shepherd

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Day 1 review

- ★ Contrast average outcome values across populations of individuals defined by covariate values, while accounting for correlation
 - Focus on a generalized linear model with regression parameters β , which characterize the systemic variation in Y across covariates X

$$y_i = \{y_{i1}, y_{i2}, \dots, y_{im_i}\}^T \quad \text{Outcomes}$$

$$x_{ij} = \{1, x_{ij1}, x_{ij2}, \dots, x_{ijp}\} \quad \text{Covariates}$$

$$X_i = \{x_{i1}, x_{i2}, \dots, x_{im_i}\}^T \quad \text{Design matrix}$$

$$\beta = \{\beta_0, \beta_1, \beta_2, \dots, \beta_p\}^T \quad \text{Regression parameters}$$

for $i = 1, \dots, n$ and $j = 1, \dots, m_i$

- Longitudinal correlation structure is a nuisance feature of the data (Liang and Zeger, 1986)

Mean model

Assumptions

- Observations are independent across subjects
- Observations may be correlated within subjects

Mean model: Primary focus of the analysis

$$\begin{aligned}E[Y_{ij} \mid x_{ij}] &= \mu_{ij} \\g(\mu_{ij}) &= x_{ij}\beta\end{aligned}$$

- May correspond to any generalized linear model with link $g(\cdot)$

Continuous outcome	Count outcome	Binary outcome
$E[Y_{ij} \mid x_{ij}] = \mu_{ij}$ $\mu_{ij} = x_{ij}\beta$	$E[Y_{ij} \mid x_{ij}] = \mu_{ij}$ $\log(\mu_{ij}) = x_{ij}\beta$	$P[Y_{ij} = 1 \mid x_{ij}] = \mu_{ij}$ $\text{logit}(\mu_{ij}) = x_{ij}\beta$

- Characterizes a **marginal** mean regression model
 - ▶ μ_{ij} does not condition on anything other than x_{ij}

Covariance model

Longitudinal correlation is a nuisance; secondary to mean model of interest

1. Assume a form for **variance** that may depend on μ_{ij}

$$\text{Continuous outcome:} \quad \text{Var}[Y_{ij} \mid x_{ij}] = \sigma^2$$

$$\text{Count outcome:} \quad \text{Var}[Y_{ij} \mid x_{ij}] = \mu_{ij}$$

$$\text{Binary outcome:} \quad \text{Var}[Y_{ij} \mid x_{ij}] = \mu_{ij}(1 - \mu_{ij})$$

which may also include a scale or dispersion parameter $\phi > 0$

2. Select a model for longitudinal **correlation** with parameters α

$$\text{Independence:} \quad \text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = 0$$

$$\text{Exchangeable:} \quad \text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = \alpha$$

$$\text{Auto-regressive:} \quad \text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = \alpha^{|j-j'|}$$

$$\text{Unstructured:} \quad \text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = \alpha_{jj'}$$

Covariance model

Longitudinal correlation is a nuisance; secondary to mean model of interest

- Assume a form for variance that depends on μ
- Select a model for longitudinal correlation with parameters α

$$\begin{aligned}\text{Var}[Y_{ij} \mid X_i] &= V(\mu_{ij}) \\ S_i(\mu_i) &= \text{diag } V(\mu_{ij})\end{aligned}$$

$$\begin{aligned}\text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] &= \rho(\alpha) \\ R_i(\alpha) &= \text{matrix } \rho(\alpha)\end{aligned}$$

$$\begin{aligned}\text{Cov}[Y_i \mid X_i] &= V_i(\beta, \alpha) \\ &= S_i^{1/2} R_i S_i^{1/2}\end{aligned}$$

Correlation models

Independence: $\text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = 0$

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Exchangeable: $\text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = \alpha$

$$\begin{bmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{bmatrix}$$

Correlation models

Auto-regressive: $\text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = \alpha^{|j-j'|}$

$$\begin{bmatrix} 1 & \alpha & \alpha^2 & \cdots & \alpha^{m-1} \\ \alpha & 1 & \alpha & \cdots & \alpha^{m-2} \\ \alpha^2 & \alpha & 1 & \cdots & \alpha^{m-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha^{m-1} & \alpha^{m-2} & \alpha^{m-3} & \cdots & 1 \end{bmatrix}$$

Unstructured: $\text{Corr}[Y_{ij}, Y_{ij'} \mid X_i] = \alpha_{jj'}$

$$\begin{bmatrix} 1 & \alpha_{21} & \alpha_{31} & \cdots & \alpha_{m1} \\ \alpha_{12} & 1 & \alpha_{32} & \cdots & \alpha_{m2} \\ \alpha_{13} & \alpha_{23} & 1 & \cdots & \alpha_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha_{1m} & \alpha_{2m} & \alpha_{3m} & \cdots & 1 \end{bmatrix}$$

Correlation models

Correlation between any two observations on the same subject. . .

- **Independence:** . . . is assumed to be zero
 - ▶ Always appropriate with use of robust variance estimator (large n)
- **Exchangeable:** . . . is assumed to be constant
 - ▶ More appropriate for clustered data
- **Auto-regressive:** . . . is assumed to depend on time or distance
 - ▶ More appropriate for equally-spaced longitudinal data
- **Unstructured:** . . . is assumed to be distinct for each pair
 - ▶ Only appropriate for short series (small m) on many subjects (large n)

Semi-parametric

- Specification of a mean model and correlation model does not identify a complete probability model for the outcomes
- The [mean, correlation] model is semi-parametric because it only specifies the first two moments of the outcomes
- Additional assumptions are required to identify a complete probability model and a corresponding parametric likelihood function (GLMM)

Question: Without a likelihood function, how do we estimate β and generate valid statistical inference, while accounting for correlation?

Answer: Construct an unbiased estimating function

Estimating functions

The estimating function for estimation of β is given by

$$\mathcal{U}_{\beta}(\beta, \alpha) = \sum_{i=1}^n D_i^{\top} V_i^{-1} (Y_i - \mu_i)$$

$$\mu_i = g^{-1}(X_i \beta)$$

$$D_i = \frac{\partial \mu_i}{\partial \beta}$$

- V_i is the 'working' variance-covariance matrix: $\text{Cov}[Y_i \mid X_i]$
 - ▶ Depends on the assumed form for the variance: $\text{Var}[Y_{ij} \mid x_{ij}]$
 - ▶ Depends on the specified correlation model: $\text{Corr}[Y_{ij}, Y_{ij'} \mid X_i]$
- V_i may also be written as a covariance weight matrix: $W_i = V_i^{-1}$
- $\mathcal{U}_{\beta}(\beta, \alpha)$ depends on the model or value for α

Generalized estimating equations

Setting an estimation function equal to 0 defines an estimating equation

$$\begin{aligned} 0 &= \mathcal{U}_\beta(\hat{\beta}, \alpha) \\ &= \sum_{i=1}^n D_i^\top V_i^{-1}(Y_i - \hat{\mu}_i) \end{aligned}$$

with $\hat{\mu}_i = g^{-1}(X_i \hat{\beta})$

- 'Generalized' because it corresponds to a GLM with link function $g(\cdot)$
- Solution to the estimation equation defines an estimator $\hat{\beta}$
- $\mathcal{U}_\beta(\hat{\beta}, \alpha)$ depends on the model or value for α
 - ▶ Moment-based estimation of α based on residuals
 - ▶ A second set of estimating equations for α

Generalized estimating equations: Intuition

$$0 = \sum_{i=1}^n \underbrace{D_i^T}_{\boxed{3}} \underbrace{V_i^{-1}}_{\boxed{2}} \underbrace{(Y_i - \hat{\mu}_i)}_{\boxed{1}}$$

- 1 The model for the mean, $\mu_i(\beta)$, is compared to the observed data, Y_i ; setting the equations to equal 0 tries to minimize the difference between **observed** and **expected**
- 2 Estimation uses the inverse of the variance (covariance) to weight the data from subject i ; more weight is given to differences between observed and expected for those subjects who contribute more information
- 3 This is simply a 'change of scale' from the scale of the mean, $\mu_i(\beta)$, to the scale of the regression coefficients (covariates)

Properties of $\hat{\beta}$

Suppose Y_i is continuous so that $E[Y_i | X_i] = X_i\beta$ and $\text{Cov}[Y_i | X_i] = V_i$

$$\hat{\beta} = \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1} \sum_{i=1}^n X_i^T V_i^{-1} Y_i$$

- $\hat{\beta}$ is **unbiased** assuming $E[Y_i | X_i] = X_i\beta$ is correct

$$\begin{aligned} E[\hat{\beta}] &= \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1} \sum_{i=1}^n X_i^T V_i^{-1} E[Y_i] \\ &= \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1} \sum_{i=1}^n X_i^T V_i^{-1} X_i \beta \\ &= \beta \end{aligned}$$

Properties of $\hat{\beta}$

- $\hat{\beta}$ is **efficient** assuming $\text{Cov}[Y_i | X_i] = V_i$ is correct

$$\begin{aligned}\text{Cov}[\hat{\beta}] &= \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1} \\ &\quad \times \left(\sum_{i=1}^n X_i^T V_i^{-1} \text{Cov}[Y_i] V_i^{-1} X_i \right) \\ &\quad \times \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1} \\ &= \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1}\end{aligned}$$

which is known as the model-based variance estimator

Properties of $\hat{\beta}$

If $\text{Cov}[Y_i | X_i] \neq V_i$, then use an empirical estimator

$$\begin{aligned}\text{Cov}[\hat{\beta}] &= \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1} \\ &\quad \times \left(\sum_{i=1}^n X_i^T V_i^{-1} (Y_i - \mu_i) (Y_i - \mu_i)^T V_i^{-1} X_i \right) \\ &\quad \times \left(\sum_{i=1}^n X_i^T V_i^{-1} X_i \right)^{-1}\end{aligned}$$

- Also known as sandwich, robust, or Huber-White variance estimator
- Requires sufficiently large sample size ($n \geq 40$)
- Requires sufficiently large sample size relative to cluster size ($n \gg m$)

$(Y_i - \mu_i)(Y_i - \mu_i)^\top$ is a poor estimate of $\text{Cov}[Y_i]$ for each i

- However, a good estimate for each i is not required
- Rather, need a good estimate of the average (total) covariance

$$B_n = \frac{1}{n} \sum_{i=1}^n D_i^\top V_i^{-1} \text{Cov}[Y_i] V_i^{-1} D_i$$

$$\hat{B}_n = \frac{1}{n} \sum_{i=1}^n D_i^\top V_i^{-1} (Y_i - \mu_i)(Y_i - \mu_i)^\top V_i^{-1} D_i$$

- $\hat{B}_n$ can be well estimated with sufficient independent replication, i.e. sufficiently large sample size relative to cluster size

Properties of $\hat{\beta}$

- $\hat{\beta}$ is a consistent estimator for β even if the model for longitudinal correlation is incorrectly specified, i.e. $\hat{\beta}$ is 'robust' to correlation model mis-specification
- However, the variance of $\hat{\beta}$ must capture the correlation in the data, either by choosing the correct correlation model, or via an alternative variance estimator
- Selecting an approximately correct correlation model will yield a more efficient estimator for β , i.e. $\hat{\beta}$ has the smallest variance (standard error) if the correlation model is correctly specified

Comments

- GEE is specified by a mean model and a correlation model
 1. A regression model for the average outcome, e.g. linear, logistic
 2. A model for longitudinal correlation, e.g. independence, exchangeable
- GEE also computes an empirical variance estimator (aka sandwich, robust, or Huber-White variance estimator)
- Empirical variance estimator provides valid standard errors for $\hat{\beta}$ even if the correlation model is incorrect, but requires $n \geq 40$ and $n \gg m$

Question: If the correlation model does not need to be correctly specified to obtain a consistent estimator for β or valid standard errors for $\hat{\beta}$, why not always use an independence working correlation model?

Answer: Selecting a non-independence or weighted correlation model

- Permits use of the model-based variance estimator
- May provide improved efficiency for $\hat{\beta}$

Variance estimators

- **Independence estimating equation:** An estimation equation with a working independence correlation model
 - ▶ Model-based standard errors are generally not valid
 - ▶ Empirical standard errors are valid given large n and $n \gg m$
- **Weighted estimation equation:** An estimation equation with a non-independence working correlation model
 - ▶ Model-based standard errors are valid if correlation model is correct
 - ▶ Empirical standard errors are valid given large n and $n \gg m$

Estimating equation	Variance estimator	
	Model-based	Empirical
Independence	–	+ / –
Weighted	– / +	+

Inference for β

Consider testing one or more parameters in nested models

$$H: \beta = \begin{bmatrix} \beta_1 \\ 0 \end{bmatrix} \quad \text{versus} \quad K: \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix},$$

i.e., $H: \beta_2 = 0$

- Wald test (based on coefficient and standard error) is generally valid
 - ▶ Requires computation under the alternative hypothesis K
- Likelihood ratio test not available; not relied on a likelihood function

Effect modification

- Association of interest varies across levels of another variable, or another variable modifies the association of the variable of interest
- Modeling of effect modification is achieved by interaction terms

$$E[Y \mid x, t] = \beta_0 + \beta_1 x + \beta_2 t + \beta_3 x \times t$$

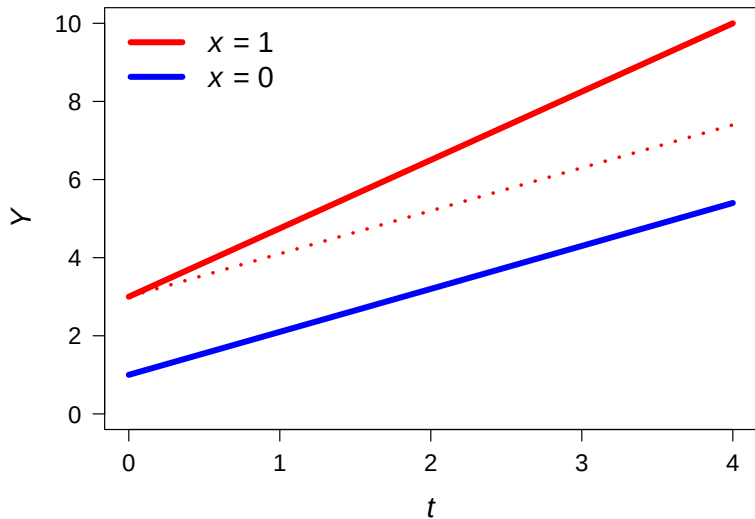
with

- ▶ A binary variable x for drug: 0 for placebo, 1 for treatment
- ▶ A continuous variable t for time since randomization
- Wish to examine whether treatment modifies the association between time since randomization and serum albumin

$$\text{Placebo: } E[Y \mid x = 0, t] = \beta_0 + \beta_2 t$$

$$\begin{aligned}\text{Treatment: } E[Y \mid x = 1, t] &= \beta_0 + \beta_1 + \beta_2 t + \beta_3 t \\ &= (\beta_0 + \beta_1) + (\beta_2 + \beta_3)t\end{aligned}$$

Effect modification



Effect modification

- Contrasts for t (time) depend on the value for x (drug)

$$\begin{aligned} & E[Y \mid x, t + 1] - E[Y \mid x, t] \\ &= \{\beta_0 + \beta_1 \cdot x + \beta_2 \cdot (t + 1) + \beta_3 \cdot x \cdot (t + 1)\} \\ &\quad - \{\beta_0 + \beta_1 \cdot x + \beta_2 \cdot t + \beta_3 \cdot x \cdot t\} \\ &= \beta_2 + \beta_3 x \end{aligned}$$

- β_2 compares the mean albumin level between two placebo-treated populations whose time since randomization differs by 1 year ($x = 0$)
- $\beta_2 + \beta_3$ compares the mean albumin level between two drug-treated populations whose time since randomization differs by 1 year ($x = 1$)
- Hence β_3 represents a difference evaluating whether the association between time and serum albumin differs between treatment groups
- A hypothesis test of $\beta_3 = 0$ can be used to evaluate the difference

Summary

- Primary focus of the analysis is a marginal mean regression model that corresponds to any GLM
- Longitudinal correlation is secondary to the mean model of interest and is treated as a nuisance feature of the data
- Requires selection of a 'working' correlation model
- Lack of a likelihood function implies that likelihood ratio test statistics are unavailable; hypothesis testing with GEE uses Wald statistics
- Working correlation model does not need to be correctly specified to obtain a consistent estimator for β or valid standard errors for $\hat{\beta}$, but efficiency gains are possible if the correlation model is correct

Issues

- Accommodates only one source of correlation: Longitudinal **or** cluster
- GEE requires that any missing data are missing completely at random
- Issues arise with time-dependent exposures and covariance weighting

Key points

- Marginal mean regression model
- Model for longitudinal correlation
- Only one source of positive or negative correlation
- Semi-parametric model: mean + correlation
- Form an unbiased estimating function
- Estimates obtained as solution to estimating equation
- Model-based or empirical variance estimator
- Robust to correlation model mis-specification
- Large sample: $n \geq 40$
- Efficiency of non-independence correlation models
- Testing with Wald tests
- Marginal or population-averaged inference
- Missing completely at random (MCAR)
- Time-dependent covariates and endogeneity
- R package geepack

Big picture

- Provide valid estimates and standard errors for regression parameters of interest even if the correlation model is incorrectly specified (+)
- Empirical variance estimator requires large sample size (−)
- Always provide population-averaged inference regardless of the outcome distribution; ignores subject-level heterogeneity (+/−)
- Accommodate only one source of correlation (−/+)
- Require that any missing data are missing completely at random (−)